

BENEDICT E. UMEH

Senior Full Stack Engineer | AI Conversational Systems Specialist
Customer Experience Technology | Data & Analytics

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PROFESSIONAL SUMMARY

Senior Full Stack Engineer with over 7 years of experience delivering production-ready, scalable applications across logistics, e-commerce, hospitality, healthcare, and fintech sectors. I specialise in building AI-powered conversational systems that transform customer experience—my work includes designing and deploying PhD-level chatbots using OpenAI GPT-4o with function calling, creating interactions that are warm, empathetic, naturally human, and commercially effective.

What sets me apart is my unique combination of deep technical engineering expertise and genuine frontline customer experience. Having worked 2+ years in hospitality operations, I understand intimately what makes customer interactions feel natural versus robotic, how to handle escalations with empathy, and how to maintain brand voice consistency across every touchpoint. This dual perspective enables me to build AI systems that don't just function correctly—they genuinely connect with customers.

My technical foundation spans the full stack: TypeScript, Python, Java, Node.js, Vue.js, React, and cloud infrastructure on AWS. I bring an MSc in Data Science, enabling a rigorous, data-driven approach to performance optimization, CSAT improvement, and continuous iteration. I thrive in fast-paced, agile environments where progress over perfection drives innovation.

TECHNICAL EXPERTISE

AI & Conversational Systems

OpenAI GPT-4o API, Prompt Engineering & Optimization, Function Calling & Tool Orchestration, Multi-Agent AI Workflows, Conversation Flow Design, AI Quality Assurance & Testing, Response Quality Optimization, Tone & Empathy Analysis, Edge Case Handling, Escalation Logic Design

Programming Languages & Frameworks

JavaScript/TypeScript (Expert), Python (Advanced), Java (Proficient), PHP (Proficient) | Node.js, Express.js, Vue.js, React 18, Spring Framework, WebSockets, REST API Design

Cloud Infrastructure & DevOps

Amazon Web Services (EC2, S3, Lambda, RDS, CloudWatch), Docker Containerization, CI/CD Pipelines (GitHub Actions, Jenkins), Infrastructure as Code, Microservices Architecture, API Gateway Management

Databases & Data Engineering

PostgreSQL, MongoDB, MySQL, Redis, Drizzle ORM, Database Optimization, Query Performance Tuning, Data Modeling, ETL Pipelines

Analytics & Business Intelligence

Power BI, Google Analytics 4, Custom Dashboard Development, A/B Testing Frameworks, KPI Tracking, Performance Metrics Analysis, CSAT/NPS Measurement

Customer Experience & Quality

Brand Voice Consistency, Service Quality Assurance, Escalation Handling, Customer Journey Optimization, Stakeholder Communication, Agile/Scrum Methodology

PROFESSIONAL EXPERIENCE

Founder & Lead Software Engineer

StackRise Ltd – Enterprise SaaS Platform | Manchester, UK | December 2024 – Present

Founded and built an enterprise-grade SaaS platform from the ground up, serving businesses with AI-powered automation, customer relationship management, and e-commerce capabilities.

- Designed and implemented STAX AI Chatbot using OpenAI GPT-4o with function calling—a PhD-level conversational assistant that handles complex customer queries with warmth, empathy, and operational precision. The chatbot was specifically engineered to never sound robotic or overly factual; every response maintains natural, human-like conversational flow while ensuring commercial effectiveness.
- Built comprehensive AI quality assurance framework: systematically reviewed voice and digital interactions for tone, empathy, brand voice consistency, and operational accuracy. Maintained a structured library of best-in-class example responses alongside documented failure cases for continuous learning and prompt refinement.

- Implemented 3 CRM integration tools through AI function calling (lead collection, callback scheduling, profile updates), ensuring all API actions complete correctly and align precisely with customer intent. Built validation systems to flag incorrect advice, failed actions, or compliance risks.
- Developed proactive engagement triggers and real-time conversation quality monitoring. Adopted a rigorous data-led approach: identified where the AI failed to handle queries, updated prompts accordingly, and measured the impact of every change against CSAT and resolution metrics.
- Architected the full-stack SaaS platform using TypeScript, Node.js, Express.js, and PostgreSQL, delivering 15+ production features with 99.9% uptime. Built RESTful API architecture with 50+ endpoints, OAuth 2.0/JWT authentication, and comprehensive error handling.
- Engineered complete e-commerce payment processing system with Stripe API integration, webhook handling, and subscription management, reducing payment completion time by 90%.
- Optimized database queries and implemented intelligent caching strategies, improving API response times by 60% (from 800ms to 300ms).

Technologies: TypeScript, Python, Node.js, PostgreSQL, OpenAI GPT-4o, Stripe, AWS S3, WebSockets, Jest, CI/CD, Docker

Guest Service Assistant

Ibis Styles Leeds City Centre Arena | Leeds, UK | September 2023 – Present

Front-desk operations role that has provided invaluable insight into real-world customer communication patterns and service excellence standards.

- Manage all night shift front-desk operations for a 200+ room hotel, delivering outstanding customer service with consistent warmth, clarity, and genuine empathy. Handle check-ins, guest inquiries, complaint resolution, and daily reporting.
- Self-initiated and built an AI Virtual Hotel Concierge using React and WebSockets, automating responses to 200+ common guest queries. This innovation reduced call volume by 35% while maintaining high guest satisfaction scores.
- Developed deep, practical understanding of customer communication patterns—specifically what makes responses feel natural versus robotic, how tone affects customer perception, and how to handle escalations effectively while maintaining brand standards.
- Gained extensive experience resolving complex customer journeys across both service and sales scenarios, maintaining consistent quality standards even under pressure.

Full Stack Developer (Contract)

Qalhata Technology – AI/Autonomous Systems | Cheshire, UK | November 2024 – Present

- Lead development initiatives for enterprise clients in the AI and autonomous systems sector, implementing end-to-end analytics pipelines and executive-level performance tracking dashboards.
- Built automated reporting infrastructure using Python and Power BI, dramatically reducing stakeholder report turnaround from 5 days to 1 day (80% improvement).
- Integrated Google Analytics 4 with custom event tracking, providing actionable insights that directly informed data-driven business decisions.
- Mentor junior developers on quality standards, code review best practices, and Agile development methodology.

Technologies: Python, JavaScript, PostgreSQL, Power BI, Google Analytics API, Vue.js

Web Developer

Deskstones Ltd | Leeds, UK | February 2024 – Present

- Led complete corporate website rebuild, achieving 45% page load improvement through systematic performance optimization, code refactoring, and database query tuning.
- Increased organic SEO visibility by 30% and conversion rate by 25% through rigorous A/B testing and user experience optimization.

Freelance Software Engineer & Founder

bhenmedia.com | Leeds, UK & Lagos, Nigeria | 2017 – Present

- Founded and operated a successful web development consultancy, delivering 50+ complete web applications for clients across e-commerce, hospitality, healthcare, and logistics sectors.
- Built e-commerce platforms with comprehensive payment processing (Stripe, PayPal), order tracking systems, and inventory management.
- Successfully migrated multiple legacy PHP applications to modern TypeScript/Node.js architectures, improving scalability and maintainability.
- Notable client projects: Ebluniversal.com, Slot.ng, Vivano.com.ng, Rockae.com

Technologies: JavaScript, TypeScript, PHP, Python, Vue.js, React, MySQL, PostgreSQL, Stripe, AWS

AI & CONVERSATIONAL SYSTEMS PROJECTS

STAX PhD-Level AI Assistant

Personal Project | December 2024

Designed and implemented an enterprise-grade AI chatbot specifically engineered for natural, human-like conversational flows. Key focus areas included:

- Ensuring responses never sound robotic, overly factual, or stiff—maintaining warmth and conversational naturalness
- Building a comprehensive conversation quality framework with systematic review of interactions for tone, empathy, and accuracy
- Implementing edge case handling, escalation logic, and stress-testing for operationally complex customer journeys
- Maintaining structured logs of issues, themes, and priorities; capturing best-in-class responses for training
- Applying data-driven methodology: identifying AI failure points, updating prompts, measuring change impact

Multi-Agent AI Systems

Research & Development | November 2024 – Present

- Engineered sophisticated multi-agent AI workflows with 4 specialized assistants, reducing user research time by 70%
- Built agentic AI systems using OpenAI function calling with multi-step tool orchestration across 6-stage analysis workflows
- Implemented validation systems to ensure API actions complete correctly and align with user intent; built compliance risk flagging

EDUCATION

Master of Science in Data Science

Leeds Beckett University, United Kingdom | 2023

Dissertation: Applied machine learning techniques to business analytics problems

Bachelor of Science in Information Technology & Business Information Systems

Middlesex University, London, United Kingdom | 2017

Advanced Diploma in Software Engineering

Aptech Computer Institute, Lagos, Nigeria | 2016

PROFESSIONAL CERTIFICATIONS

- IBM Data Analyst Professional Certificate
- Google Data Analytics Professional Certificate
- Google Analytics IQ Certification
- AWS Cloud Practitioner (In Progress)

KEY ACHIEVEMENTS

- Built production AI chatbot with natural, empathetic responses—proactive customer engagement across voice and digital
- Reduced hotel call volume by 35% through AI-powered virtual concierge with quality conversation design
- Optimized API response times by 60% through data-driven performance analysis and intelligent caching
- Delivered 50+ applications across e-commerce, logistics, healthcare, and fintech with consistent quality
- Deep frontline customer experience—understanding what makes interactions feel genuinely human and on-brand